CS 6012: Autumn 2026 — Open Project

Intermediate Checkpoint #2: Status Report (Submission 2: Code-Based Project)

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1. Title of the Project

GNN Molecular Substructure Search: Accuracy vs. Speed Trade-offs Against Classical Fingerprint Screening

2. Problem Statement

Exact molecular substructure search via subgraph isomorphism algorithms (e.g., RDKit's Ullmann/VF2-based HasSubstructMatch) provides reliable ground truth but scales poorly across large chemical databases due to combinatorial complexity. We train Graph Neural Networks (GNNs) to act as an ultrafast approximate candidate filter predicting whether a query fragment Q is contained within target molecule T. The primary objective is to investigate the accuracy-speed Pareto frontier when pairing a learned GNN shortlist with exact verification, benchmarked directly against industrial cheminformatics standards (RDKit Chem.PatternFingerprint).

3. Extensions to Prior Work [IN PROGRESS]

- **NeuroMatch-Style Order Embeddings:** Extended beyond standard pairwise binary classifiers by mapping molecular subgraphs to a learned non-negative cone ($z_Q \leq z_T$ element-wise) optimized via max-margin violation loss ($E(Q, T) = ||max(0, z_Q - z_T)||^2$). This enables precomputing target database embeddings once, reducing search across 50,000+ molecules to instantaneous vectorized matrix subtraction.
- **Bond-Aware Chemical Graph Encoding:** Replaced generic graphs with chemistry-aware GINEConv message-passing layers utilizing 10-dimensional one-hot atom features and 4-dimensional bond type features matched precisely to query SMARTS semantics.
- **Two-Stage Cascaded Screening:** Formulated a cascaded retrieval architecture where the order-embedding GNN shortlists candidate molecules at a calibrated ≥98% recall threshold, followed by RDKit exact verification solely on the shortlist.

4. How Do We Feel? [In-the-zone]

We have implemented the full end-to-end pipeline in PyTorch Geometric and RDKit with zero syntax or runtime errors, configured GPU CUDA acceleration, and completed an empirical pilot benchmark on a 5,000-molecule database. Both teammates have established active contributions.

5. Contributions of Members

- **Aayush Gupta (25m0140):** Implemented PyG graph featurization (atom/bond one-hot encodings), GINEConv Siamese encoder, Model A (BCE) and Model B (Order Embedding) loss functions, decision threshold optimization, and Colab integration.
- **Srijal Gupta (25m0387):** Built ZINC250k data loading and partitioning, subgraph query generation (4–12 atoms + 20 functional groups), RDKit ground-truth labeling, Pattern Fingerprint screening baseline, and comparative timing/profiling benchmarks.
- **Joint Deliverables:** Cascaded search formulation, error analysis by query size, report documentation, and presentation recording.

6. Repository & Updated Schedule

- **GitHub Repository:** https://github.com/srijal20/GNN-Substructure-Search-project
  (Active commits by both members; contains full source code, pilot CSV benchmarks, tradeoff plots, and Colab notebook)
- **Updated Schedule Spreadsheet:**
  https://docs.google.com/spreadsheets/d/1264hAmtv7T5bGPyuDbSqR52UGxm6RZCEa7NFxWk-cQ4/edit?usp=sharing